generality we can take p_t as one because we further assume super neutrality of money (increasing money supply doesn't have impact on prices)

The profit maximizing expression of representative firm is

$$\pi(t) = F(.) - w(t)L(t) - [r(t) + \delta]k(t) \qquad \ldots(24.34)$$

The 1st order condition for optimality

$$\frac{\partial \pi(t)}{\partial L(t)} = 0 \Rightarrow \frac{\partial F(.)}{\partial L(t)} = w(t) \qquad \ldots(24.35)$$

$$\frac{\partial \pi(t)}{\partial k(t)} = 0 \Rightarrow \frac{\partial F(.)}{\partial k(t)} = r(t) + \delta \qquad \ldots(24.36)$$

Equation (24.35) and (24.36) in light of competitive market, the return on production inputs is equal to the corresponding marginal productivity.

We can derive resource accumulation constraint from household budget constraint. The agent/household accumulates wealth or capital as the difference between received income (from labor and capital less consumption $(c(t) \geq 0$, represents the level of consumption)

$$K'(t) = w(t)\,L(t) + r(t)\,k(t) - c(t), K(0) \qquad \ldots(24.34)$$

In intensive form equation

$$k' = w(t) + r(t)k(t) - c(t)k(0)\,\text{given} \qquad \ldots (24.38)$$

With $c(t) \cong \dfrac{c(t)}{L(t)}$ and $k = \dfrac{k}{L}$, $y = \dfrac{y(t)}{L(t)}$

It we take production function as Cobb-Douglas production function $y(t) = F(K_t, L_t) = A K_t^{\alpha} L^{1-\alpha}$ and calculated marginal productivity labor and marginal productivity of capital, then we get

$$\frac{\partial F(.)}{\partial L(t)} = (1-\alpha)A\,k(t)^{\alpha} = w_t \qquad \ldots (24.39)$$

$$\frac{\partial F(.)}{\partial k(t)} = \alpha\,A k(t)^{1-\alpha} = r(t) + \delta \qquad \ldots (24.40)$$

The sum of two return on two inputs is equal to the total output produced in the economy.

$$y(t) = w(t) + [r(t) + \delta]k(t) = (1-\alpha)Ak(t)^{\alpha} + \alpha A\,k(t)^{\alpha} = Ak(t)^{\alpha} \qquad \ldots (24.40)$$

Rewriting equation (24.38) we get

$k'(t) = w(t) + (r(t) + \delta)\,k(t) - c(t), -\delta k(t)\,k(0),$ given in intensive form we get

$$k' = A k^{\alpha}(t)^{\alpha} - c(t) - \delta k(t), k(0)\,\text{given} \qquad \ldots (24.41)$$

(by using 24.40) in above equation)

Differential equation (24.41) characterizes the process of capital accumulation in an economy: it states that net capital accumulation is the difference between gross investment (output less consumption) and capital depreciation Equation (24.41) constitutes the basic relation through which we discuss economic growth.

Now we consider the preference of household which is expressed as utility function. Assume that consumption is the only argument of such function and it is expressed as $u(c(t))$ for some point of time t.

The utility function has few basic standard properties namely continuity differentiability and concavity. Function u is both increasing $(u' > 0)$ and concave $(u'' < 0)$ mean that marginal utility of consumption is positive and diminishing (Gossen's first law), additional increments in consumption will generate progressively lower increases in utility.

A class of utility function which obeys above mentioned properties is the constant elasticity of intertemporal substitution (CEIS) function A function of this class is typically formulated as

$$u(c(t)) = \frac{c_t^{1-\theta} - 1}{1-\theta}, \theta \in (0, \propto)/\{1\} \qquad \dots (24.42)$$

Here θ is the coefficient of relative risk aversion defined as $-c_t u''(c(t))/u'(c(t))$. The coefficient of relative risk aversion tells us how the household is going to rank different lotteries.

$-\theta$ plays another role: It is also the inverse of the instantaneous elasticity of intertemporal substitution (EIS). The intertemporal elasticity of substitution between two points in time t and s > t is defined as

$\sigma_{t,s} = -dl_n(c_s / c_t) / d\ln(u(c_s)/u(c_t))$ if we take limit of this expression $s \to t$ we yet

$$\sigma = -u'(c_t) / c_t u''(c_t) = \frac{1}{\theta}$$

The EIS tells us how willing the household is to shift consumption from one period to another.

The case $\theta = 1$ is a particular case, obtained by applying L'-Hopital's rule to above expression (to equation 24.42) which allows us defining utility as a logarithmic function

$$u(t) = \ln(c_t) \qquad \dots (24.44)$$

The optimal growth problem is to maximize the following

$$\underset{\max}{u} = \int_0^{\propto} e^{-\rho t} u[c(t)] dt \qquad \dots (24.45)$$

where $\rho \geq 0$ represents the rate of time preference.

Mathematically speaking we have a objective function written in equation (24.45) which we wish to maximize subject to constrain which is in equation (24.41). Further $c(t)$ is control variable and *k*(t) is state variable.

The agent chooses the trajectory of consumption that allows for the highest intertemporal utility.

The Ramsay model (in some literature Ramsey-Cass–Koopman model) is different from Solow model – in contrast to Solow model in which savings are exogenous; in Ramsey model savings are optimally derived given intertemporal preference of representative agent; thus, dynamic inefficiencies are ruled out), and is a constrained continuous dynamic optimisation problem.

(Please refer to discounted problem in Unit: 23 for formula)

Forming Hamiltonian functional we get

$$H^* = u[c(t)] + \lambda_t[A k^{\alpha}(t) - c(t) - \delta k(t)] \qquad \ldots (24.46)$$

where λ_t is co-state variable associated with state variable, *k*. For us, $u[c(t)]$ is from equation (24.42).Applying necessary condition for optimisation, we get

$$\frac{\partial H^*}{\partial c(t)} = 0 \Rightarrow c(t)^{-\theta} = \lambda_t \qquad \ldots (24.43)$$

From co-state equation, we get

$$\lambda_t' = \rho\lambda_t - \frac{\partial H^*}{\partial k_t} \Rightarrow \lambda' t = [\rho + \delta - \alpha A k(t)^{-(1-\alpha)}]\lambda_t \qquad \ldots (24.44)$$

In equation $\theta > 0$ is the concavity parameter of the utility function

Differentiating equation (24.43) and using that in equation (24.44) we get

$$c'(t) = \frac{1}{\theta}[\alpha A k(t)^{-(1-\alpha)} - (\rho + \delta)]c(t) \qquad \ldots (24.45)$$

Transversality condition is essential for solution i.e.,

$$\lim_{t\to\infty} \lambda_t e^{-\rho t} k(T) = 0$$

The dynamics of Ramsey model is found by analysing (24.41) and equation (24.45)

In order to derive steady-state locus we need to set $k'(t) = 0$ and $c'(t) = 0$ which in turn determine c^* and k^*

$$k^* = \left(\frac{\alpha A}{\rho + \delta}\right)^{1/1-\alpha} \qquad \ldots (24.46)$$

$$C^* = \left(\frac{1}{\alpha}\rho + \frac{1-\alpha}{\alpha}\delta\right)\left(\frac{\alpha A}{\rho + \delta}\right)^{1/(1-\alpha)} \qquad \ldots (24.47)$$

The equilibrium level of capital and consumption is directly proportional to value of the technology index. The rate of time preference and depreciation rate have inverse proportional relation on capital accumulation, the impact of these parameters on consumption is not straight forward unless we attribute exact values of various involved parameter.

Steady state levels of income and saving can be obtained from equation (24.46) and (24.47)

$$y^* = A(k^*)^\alpha = A^{1/(1-\alpha)}\left(\frac{\alpha}{\rho+\delta}\right)^{\frac{\alpha}{1-\alpha}} \quad \dots (24.48)$$

$$s^* = y^* - c^* = \frac{\alpha\delta}{\rho+\delta} A^{1/(1-\alpha)}\left(\frac{\alpha}{\rho+\delta}\right)^{\frac{\alpha}{1-\alpha}} \quad \dots (24.49)$$

Note s^* is positive after consumption is maximized.

The savings are compatible with the golden rule of capital accumulation because it is being derived optimally. The law of welfare is established by the fact that the saving cannot be altered without penalizing agent's utility. (In other way any other saving rate is sub optimal).

24.9 NEOCLASSICAL THEORY OF INVESTMENT WITH ADJUSTMENT COSTS

The new Palgrave dictionary to economics states, "Given" their necessary simplicity, we often find prediction of theoretical economies we are able to analyse - are too stark relative to behaviour observed in actual economies. Thus, in a variety of settings we have adopted adjustment costs in our economic laboratories to summarize omitted frictional elements that reduce, delay or protract changes in the demand and supply of final goods and their factor inputs in response to changes in economic condition hence", hence the demand for adjustment cost model.

In vast array to macroeconomics literature, investment adjustment cost model (hence for sake of simplicity will be written adjustment cost model) explain the rate at which the capital stock catches up with its desired level. The term "adjustment" can be thought of as the additional cost incurred by firm for turning investment into production capital. The adjustment can be gradual or instantaneous and that depends on the typical mathematical model. Adjustment cost can include, installation cost, reorganization of plant, retraining of worker delivery lags, holding inventories due to tax consideration. These costs of adjustment are generally assumed as a function of investment and capital stock level.

If the desired level of capital stock is k^* and current level of capital stock is k, then the investment rate is a nonnegative function of difference between k^* and k. In order to determine optimal path of adjustment, if we assume convex adjustment cost-which increases at an increasing rate with I_t – then we can assume smooth and gradual path of adjustment, on the contrary a non-convex adjustment cost function depict 'lumpy' investment: large infrequent adjustment of capital stock.

Adjustment cost model originates from the seminal work of Eisner and Strotz (1963). We consider a model of adjustment cost of a firm, which is price taking and producing at constant return to scale. The adjustment cost function is assumed to be convex.

The firm faces following optimisation problem

$$\max V(.) = \int_0^{\infty} e^{-rt} R(t) dt$$

… (24.50)

[where $R(t) = pF(k(t), L(t)) = G(I(t), k(t)) - w(L)L(t) - p^k I$

… (24.51)

where p^k relative price of capital

$F(k(t), L(t)$ is output & w(t) is wage

Because $G(I(t) k(t))$ the adjustment cost function is convex $\frac{\partial G}{\partial I} > 0 \frac{\partial^2 G}{\partial I^2} > 0, \frac{\partial G}{\partial k} \le 0, G(0, k) = 0$

$I_{max} < \infty, I_{min} \ge 0, \delta > 0$ and $r > 0$ are depreciation rate and interest rate respectively.

Subject to $\frac{dk(L)}{dt} = I(t) - \delta k(t)$ … (24.52)

(Please refer to discounted problem in unit: 23 for formula)

Using equation (24.50) and (24.52) we formulate Hamiltonian functional as under

$$H^* = pF(k, L) - G(I, k) - wL - p^k I + \mu_t (I - \delta k)$$

… (24.52a)

(Note we are omitting 't' subscript for neatness)

The necessary condition for optimisation)

$$\frac{\partial H^*}{\partial L} = 0 \Rightarrow \frac{p \partial F(.)}{\partial L} = w \Rightarrow \frac{\partial F(.)}{\partial L} = \frac{w}{p}$$

… (24.53)

$$\frac{\partial H^*}{\partial I} = 0 \Rightarrow \frac{-\partial G(.)}{\partial I} - p^k + \mu_t = 0 \Rightarrow \mu_t = p^k + \frac{\partial G}{\partial I}$$ …

(24.54)

From co-state equation

$$\mu_t' = r\mu_t - p\frac{\partial F(.)}{\partial k} + \frac{\partial G(.)}{\partial k} + \mu_t \delta$$

$$\Rightarrow \mu'(t) = (r + \delta)\mu_t - p\frac{\partial F(.)}{\partial k} + \frac{\partial G(.)}{\partial k}$$ … (24.55)

Using (24.54) in (24.55) we get

$$\mu'(t) = (r+\delta)\left(p^k + \frac{\partial G}{\partial I}\right) - p\frac{\partial F(.)}{\partial k} + \frac{\partial G(.)}{\partial k} \qquad \ldots (24.56)$$

Before proceeding for analysis of optimal solution we have following observation

i) If we ignore adjustment cost function in equation (24.52a) then the equation becomes (24.29). In other words, without adjustment cost the model would be practically equivalent to the Jorgenson's formulation thus yielding condition analogous to equation (24.33)

ii) Because the adjustment cost function in Hamiltonian formulation is proceed with minus sigh and function is convex function, $\max - G(.) = \min G(.)$

Maximizing H^* will automatically minimizes $-G(.)$ i.e., maximizing H^* will give a minimum adjustment cost there by a maximum capital k^* that would in turn maximizes profit.

Analysis of Optimal Path

We have two equations; i.e., equation (24.52) and equation(24.56) for analysis.

a) μ is the marginal value of installed capital μ can be interpreted as the present value of the increase in net revenues that would be provided by additional unit of capital, after other inputs have been adjusted μ is always positive because p^k and $\frac{\partial G}{\partial I}$ is always greater than zero.

b) Because of assumption of constant return to scale $\frac{\partial F(.)}{\partial k}$ is constant considering $\frac{\partial G}{\partial k} < 0$ we take from equation (24.56) that μ' is downward slopping against increasing capital

c) Setting equation (24.52) to zero

$$k'(t) = I(t) - \delta k(t) = 0 \Rightarrow k^*(t) = \frac{I(t)}{\delta} \qquad \ldots (24.57)$$

Setting equation (24.56) to zero

$$0 = (r+\delta)\mu_t - p\frac{\partial F(.)}{\partial k} + \frac{\partial G(.)}{\partial k}$$

taking $r + \delta = c_1$, constant $\frac{p\partial F(.)}{\partial k} = c_2$, a constant.

$$\Rightarrow \mu_t^* = \frac{c_2 - \frac{\partial G(.)}{\partial k}}{c_1} \qquad \ldots (24.58)$$

d) As $G(.)$ is convex, assuming economy consists of fixed number but large identical firms, eq (24.56) implicitly defines the following aggregate investment function

$$I = F(\mu, p^k, k) \text{ with } \frac{\partial F(.)}{\partial \mu} > 0, \frac{\partial F(.)}{\partial k} > 0, \frac{\partial F(.)}{\partial p^k} < 0 \qquad \dots (24.59)$$

This is the general result of the model with convex and symmetric adjustment cost (Ceteris paribus, the adjustment cost of increasing the capital stock by some amount is equal to the adjustment cost of decreasing it by same amount.

e) The model shows a negative elasticity of investment to interest rate $\left(\frac{\partial I}{\partial r} < 0\right)$ - a standard characteristic of neoclassical model). From equation (24.56) a lower interest rate increases the present value of an additional capital μ resulting in a faster speed of convergence and higher investment rate.

f) Optimal k^* and μ^* is shown in figure (6)

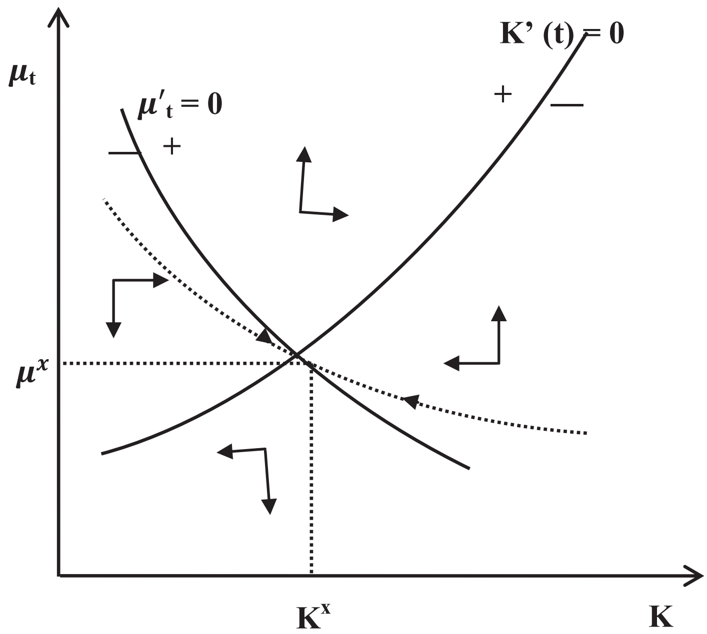

Fig. 24.6: Dynamics of μ_t and k(t)

Check Your Progress 2

1) Consider a competitive firm producing good y. The production function of the firm is given by $Y(t) = AF(k(t))$ where A is total factor productivity and k is the capital stock. The market price of capital stock and market price of output is unity. If the gross investment I is equal to $k'(t) + \delta k(t)$ where δ is depreciation rate. Assuming the instantaneous cost of gross investment for firm is equal to $I(t) + \psi(I(t))$ where ψ is a convex function for which it holds $\psi(0) = 0$ $\psi' > 0$ and $\psi'' > 0$. If the firm chooses the investment path that maximizes the present value of current and future profits. Find necessary condition for profit optimisation assuming infinite time horizon and its economic interpretation and r is real interest rate.

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24.10 LET US SUM UP

The core of dynamic analysis of economy is to bring forth every variable to its present value and then applying consumption optimisation. Beginning with simple example of monopoly, 3 examples of environmental economics are given. Note that number of co-state variables $("\lambda_t")$ always equates to the number of state dynamic equations. A detailed derivation of famous Ramsey model is provided that will be handy reading MEC-102. A general theory that includes any type of adjustment in investment done by a firm is discussed and analysed. Further we have shown that if we remove adjustment cost function then the model will become neoclassical investment theory without adjustment cost (unit 24.7)

24.11 KEY WORDS

Cost of per unit investment : A firm not only borrows for investment but also invests a part/whole of profit as investment. Further not all borrowing rate are same. Based on this consideration, cost of per unit investment is aggregate cost per unit investment incurred by firm while production. It may or may not include adjustment cost.

Logistic growth function : A function which is tangent of flattened 'S' shape curve. Characterized by $g(x_t) = \gamma x_t \left(1 - \frac{x_t}{k}\right)$ where k is carrying capacity of environment, γ is intrinsic growth rate. The stock x_t goes to k when $g(x_t) = 0$

Monopoly : A firm having exclusive control over commercial activity:

Non-renewable resources : A category of natural resources that deplete with harvesting/extraction and ultimately get exhausted example: oil, coal)

Optimal saving rate (Ramsey Model) : (Ramsey models) The steady state saving rate when consumption is optimised.

Renewable resources : A category of natural resources that regenerate according to population biology – The Quantity of harvestable stock changes due to intrinsic growth or human intervention. (Example: Fish, Tree, Shrimp,)

Transversality condition : The boundary value condition $\lim_{L \to \infty} x_t \lambda_t = 0$ which is necessary for models in which optimisation model duration is from 0 to ∞

24.12 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Progress 1

1. a) Because P_t is always restricted in sign i.e., $p_t > 0$ we need $\lim_{t\to\infty} e^{-rt}\lambda_t = 0$

b) We define Hamiltonian functional

$$H^*(.) = u(x_t) + \lambda_t P_t'$$

$$\Rightarrow H^*(.) = l_n(x_t) + \lambda_t(a + b p_t - x_t)$$

The necessary condition for optimisation

$$\frac{\partial H^*(.)}{\partial x_t} = 0 \Rightarrow \frac{1}{x_t} - \lambda_t = 0 \Rightarrow \lambda_t = \frac{1}{x_t} \qquad \dots \qquad \text{(i)}$$

From co-state condition of optimisation

$$\lambda_t' = \gamma\, \lambda_t - \frac{\partial H^*(.)}{\partial p_t} = r\lambda_t - \lambda_t b \qquad \dots \text{(ii)}$$

Using eqn. (i) in eqn. (ii) we get

$$\lambda_t' = \frac{-x_t'}{x_t^2} = r.\frac{1}{x_t} - \frac{1}{x_t}\ b$$

$$\Rightarrow \frac{X_t'}{X_t} = b - r$$

$$\Rightarrow \frac{dx_t}{X_t} = (b - r)dt$$

Integrating above equation we get,

$$\int \frac{dx_t}{x_t} = \int (b - r)\ dt$$

$$\Rightarrow l_n x_t = (r - b)t$$

$$\Rightarrow X_t^* = k\, e^{(r-b)t}$$, where k is constant. (Ans.)

Check Progress 2

1. The present value of profit of firm (objective functional) is,

$$v(.) = \int_{t=0}^{\infty} e^{-rt}\ [y(t) - I(t) - \psi(I(t)]dt$$

Therefore the current value of Hamiltonian

$$H^*(.) = [AF(k(t) - I(t) - \psi(I(t) + \lambda_t(I(t) - \delta k(t))]$$

where λ_t is the multiplier of capital accumulation constraint and represents the shadow value of an additional unit of capital at instant t (Note I(t) is control variable and k(t) is state variable)

The necessary condition for optimisation is,

$$\frac{\partial H^*(.)}{\partial I} = 0 \Rightarrow I + \psi'(I(t)) = \lambda_t \quad \text{.... (24.57)}$$

$$\lambda_t' = r\lambda_t - \frac{\partial H^*(.)}{\partial k(t)} = r\lambda_t - \frac{\partial AF(k(t))}{\partial k(t)} + \delta\lambda_t$$

$$\Rightarrow \left[(r+\delta) - \frac{\lambda'(t)}{\lambda(t)}\right]\lambda(t) = AF_k(k(t)) \quad \text{... (24.58)}$$

Equation (24.57) determines the optimal shadow value of an additional unit of capital λ as equal to marginal cost of investment. This is equal to the purchase price of capital goods (assumed equal to unity) plus the marginal installation (adjustment cost $\psi'(I(t))$.

Equation (24.58) requires that at the optimum, the firm will invest until the user cost of capital (on the left hand side) is equal to the marginal product of capital (on the right-hand side). The user cost of capital is the real interest rate, plus depreciation rate, minus the expected appreciation rate of capital stock, all multiplied by the shadow value of capital $\lambda(t)$

24.13 EXERCISES

Q1. Write profit optimisation condition for a firm in Ramsey model.

Ans. Refer Section 24.8

Q2. Explain constant elasticity of intertemporal substitution (EIS) utility function

Ans. Refer Section 24.8

Q3. Derive optimal saving rate under Ramsey model.

Ans. Refer Section 24.8

Q4. Explain the relevance of investment adjustment cost in macro models. Why the adjustment cost is assumed to be convex?

Ans. Refer Section 24.9

The convex functions have unique property which is local minimum is global minimum and this property helps us in unique convergence of dynamic optimisation.